

Exploring the technological experience and impact assessment of Augmented Reality in outdoor cultural heritage guided tours: A case study of the Chiayi Sawmill

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ABSTRACT

This study explores the application of Augmented Reality (AR) in enhancing navigation and visitor engagement at the Chiayi Sawmill in Taiwan. Using the Extended Technology Acceptance Model (ETAM), which integrates TAM2 and UTAUT, the study assesses user acceptance and factors influencing AR navigation adoption in outdoor cultural heritage sites. Data were collected from 80 visitors using AR-guided tours. Results indicate that perceived playfulness strongly affects behavioral intention, while perceived usefulness influences user attitudes, showing that visitors value both the informative content and the interactive experience. A positive user experience was found to enhance perceived ease of use, which in turn promotes acceptance of the AR system. However, the study also reveals that enjoyment of the AR content is more dependent on the system's design than on the user's prior experience. The findings validate the ETAM in outdoor heritage navigation, offering practical insights for AR application in cultural heritage, emphasizing the importance of optimizing user experience, system design, and playfulness to increase visitor satisfaction and engagement. This study contributes to extending AR research from indoor settings to outdoor cultural sites and provides a reference for future AR navigation design in cultural heritage settings.

1. Introduction

With rapid technological advancements, Augmented Reality (AR) is increasingly integrated into various fields, notably in applications such as the preservation, management, and guided tours of cultural heritage [1]. However, most studies have predominantly focused on neighboring European countries, with relatively less attention given to other countries rich in cultural heritage [2]. Cultural heritage serves not only as a testament to history but also as a crucial medium for cultural inheritance. Traditional methods of guiding visitors through cultural heritage sites, which primarily rely on static displays and oral explanations, often struggle to capture and hold visitor interest. AR, through its integration of virtual and real-world elements, offers more interactive and immersive guided tour experiences, effectively enhancing public interest in cultural heritage and bridging the gap between traditional narratives and contemporary technology [3]. This presents new opportunities for the display and educational value of cultural heritage [4,5], while also enriching public understanding and experiential depth of history and

culture [6,7].

The Chiayi Sawmill in Alishan Forestry Village, Chiayi, Taiwan (Chiayi Sawmill), is a prime example of an outdoor cultural site in Taiwan whose appeal is limited by traditional tour methods. As such, it serves as an ideal experimental site for this study. This research aims to explore the potential of AR in enhancing the tour experience at the Chiayi Sawmill, thereby increasing visitor engagement and satisfaction. Through an analysis of visitor acceptance of the AR guided tour system, the study seeks to investigate its application prospects in cultural heritage guidance.

The theoretical foundation of this study is rooted in the Technology Acceptance Model (TAM). While TAM holds a significant position in explaining technological use behavior, numerous studies indicate its limitations in fully comprehending user behaviors and decision-making processes, particularly within complex AR application environments [8]. Past research has predominantly focused on TAM and the Unified Theory of Acceptance and Use of Technology (UTAUT), yet these models exhibit certain limitations when discussing the characteristics of AR and

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user behavior. For instance, variables such as Perceived Playfulness (PP) and User Experience (UE) have been given insufficient consideration, leading to an incomplete picture of user acceptance for AR-guided tour systems.

To address these limitations, this study draws upon the literature to integrate elements of TAM2 and UTAUT, thereby developing an Extended Technology Acceptance Model (ETAM). The proposed ETAM combines key perceptual variables to comprehensively explain user behavior. Specifically, this study introduces the critical variables of Perceived Playfulness and User Experience, which have been overlooked in prior research but significantly impact the acceptance of AR technology. The proposed ETAM is designed to address the limitations of the TAM and UTAUT models, particularly within the specific context of outdoor cultural heritage tours.

By integrating and analyzing visitor acceptance of AR guided tour systems, this study validates the applicability of the ETAM, providing a more comprehensive theoretical framework to explain and predict user behavior. This research not only offers new insights into the application of AR in cultural heritage guidance but also establishes a solid theoretical foundation for future studies. Specific research objectives include: (1) Establishing an ETAM applicable to cultural heritage guidance. (2) Assessing the effectiveness of AR technology in enhancing cultural heritage tour experiences. (3) Exploring the key factors influencing visitor acceptance of AR guided tour systems. To achieve these research objectives, this study poses the following research questions: (1) How effective is the ETAM in explaining visitor behavioral intentions? (2) Can AR technology significantly enhance the tour experience at the Chiayi Sawmill? (3) What factors influence visitor acceptance of AR guided tour systems?

2. Literature review

2.1. Current application of AR technology in cultural heritage tours

AR technology has reached a considerable level of maturity, augmenting the user's perception of reality by overlaying digital information onto the physical world [1]. By allowing users to freely move around and view their surroundings, AR can facilitate interactive tours directly on personal mobile devices. This facilitates a more streamlined and accessible personal tour experience [9]. AR further provides various visual aids [4], offering interactive features that transform traditional tour formats [10]. In the field of cultural heritage management and protection, AR has shown significant effectiveness [1,11]. This includes improving exhibition content, expanding exploration, reconstruction, and virtual museum applications, enhancing cultural preservation awareness and furthering the goals of cultural education and dissemination [7]. It also reduces damage to on-site artifacts and sites, alleviates space congestion issues, and aids in the digitization of cultural heritage, addressing key shortcomings in past cultural heritage preservation and protection efforts [12].

In terms of applications for heritage tours, AR can enhance edutainment and immersion, breaking free from the limitations of traditional tours [5]. With its interactive and gamified features, AR deepens users' acquisition of tour information [13,14]. It helps generate virtual interactive content during the tour, fostering exploration, discovery, and manipulation, thus achieving knowledge creation and new insights. Not only does AR enrich the dissemination, communication, and guiding of cultural heritage narratives, but it is also gradually becoming a globally recognized trend [4,5], gaining usage in museums, art galleries, and other related exhibition spaces [6].

However, the application of this technology in outdoor cultural heritage settings still faces certain challenges in providing users with a satisfactory tour and operational experience [15]. Effective AR implementation in digital tours often adopts an engaging, sometimes playful, communication style that enhances the experience without letting entertainment overshadow the core content [16]. This approach avoids

causing users to lose focus on the tour content [1,11], which would undermine the tool's purpose of knowledge transmission and advocacy [17]. Therefore, in addition to containing essential knowledge, the tour content must also offer engaging exploration and learning opportunities, which users may find important [5]. Moreover, technical aspects such as the instability of outdoor lighting [18] and lower comfort levels [19] can also affect users' tour experience. Thus, while AR can enrich the content of outdoor cultural heritage tours, whether AR can meet user expectations for usability and performance in outdoor environments, as effectively as it does indoors, remains a key question for this study.

2.2. Progress and challenges in evaluating technology acceptance models

TAM primarily predicts behavioral outcomes based on PEOU and Perceived Usefulness (PU) [20]. This model is praised for its simplicity and clarity within the IT field, but its limitation lies in overly simplifying the technology acceptance process by not adequately considering external environmental factors, which restricts its ability to explain and predict technology acceptance behaviors [8,21]. Additionally, factors such as social values and user differences can indirectly shift the research focus, making it less suitable for evaluations in real-world environments [8,22]. Furthermore, while PU and PEOU have significant relationships with Attitude Toward Use (ATU), the correlation coefficient for PEOU is relatively low, limiting its explanatory power for ATU [23,24]. This aligns with some studies suggesting that TAM cannot fully explain user behavior [8]. Researchers suggest that considering different environmental characteristics and expanding various perceptual dimensions can improve the model's evaluation limitations [25].

To address these shortcomings, Venkatesh and Davis [22] proposed TAM2, building on the original TAM by incorporating Social Influence Processes (such as subjective norms, voluntariness, and image) and Cognitive Instrumental Processes (such as job relevance, output quality, result demonstrability, and perceived ease of use) to explain the primary drivers of PU and PEOU. TAM2 includes a social dimension to understand users' ATU and Behavioral Intention (BI), emphasizing key aspects such as social influence, facilitating conditions, and individual factors like self-efficacy [26].

Tang and Chen [24] argue that TAM2 still faces issues with explanatory power. While PU and PEOU have significant relationships with ATU, the correlation coefficient for PEOU is relatively low, limiting its explanatory power for ATU and making it less comprehensive than the UTAUT proposed by Venkatesh et al. [27]. The UTAUT primarily uses Performance Expectancy (PE), Effort Expectancy (EE), Social Influence (SI), and Facilitating Conditions (FC) as main variables, and also includes moderating variables such as gender, age, and experience. This effectively addresses the limitations of the original model's core variables, adding social and environmental factors, and enabling the technology acceptance model to adapt to various users and contexts. Among these, EE and BI are identified as the best predictors within the model [28].

Although TAM2 and UTAUT provide robust frameworks for explaining technology acceptance behaviors, the evolving nature of technology and user needs suggests that user acceptance is also influenced by additional factors, such as information quality, system quality, usage cost, personal innovativeness, risk, and facilitating conditions [29]. To address these challenges, scholars recommend incorporating more user-centric features into the models, including user experience, various perceptual needs, and usage intentions, to enhance the explanatory and predictive power of the models in different contexts [8,21].

2.3. ETAM: Evaluating technology acceptance of AR in outdoor tours

According to Pascoal et al. [30], technology acceptance models are suitable for evaluating digital tools like AR, effectively assessing PU, PEOU and ATU. Kim and Shim [31] conducted a modified TAM study on AR as an educational tool, and the results showed that TAM is highly

applicable in evaluating AR acceptance. Therefore, TAM provides a foundational framework for research evaluation. However, considering the different tools and usage contexts, many revised models have been proposed. This reflects the practical limitations of TAM but also demonstrates the potential for model modifications and extensions [21,22].

TAM2 emphasizes PU and considers the social and cognitive aspects influencing subjective norms and image, with cognitive influences related to specific technological benefits [22]. UTAUT combines elements of TAM and TAM2 to predict technology acceptance and usage [27]. This allows researchers to make more nuanced discoveries regarding usage and acceptance, particularly highlighting PEOU's significant role in determining whether users adopt new systems [32].

Building on the previously discussed strengths and weaknesses of TAM2 and UTAUT, this study synthesizes key features from both frameworks to propose the ETAM. It considers the technological usability characteristics of AR in this context, exploring participants' perceptions and evaluations when using AR tours. ETAM incorporates four key variables: User Experience (UE), System Design (SD), Perceived Playfulness (PP), and Behavioral Intention (BI). Fig. 1 illustrates how ETAM extracts suitable variables from TAM2 and UTAUT frameworks to serve as the technology evaluation model for AR outdoor tours.

- (1) UE: The quality of technology use and interaction focuses not only on human needs but also transcends the functional aspects of tools, addressing non-functional needs such as diversity and aesthetics [33]. In this context, user intentions are key drivers of technology usage, and these intentions may change as users accumulate experience [34]. Research indicates that experienced users tend to be more goal-oriented, while novices are more inclined to explore. Therefore, this study posits that incorporating user experience into TAM will help to comprehensively understand how users with varying levels of experience evaluate and accept new technologies [35]. This not only enriches TAM but also effectively addresses its existing limitations, making it better suited to the current dynamic and diverse technological environment.
- (2) PP: Davis et al. [36] introduced PP as a new influencing factor and found that both PU and PP have significant positive impacts on ATU. Additionally, Agarwal and Karahanna [37] noted that this influence is similar to the Flow Experience, playing a core role in technology use and shaping perspectives and cognitions related to technology usage. Therefore, incorporating PP into the evaluation provides a more comprehensive framework for understanding users' motivations for accepting and using new technologies. It also highlights the importance of considering

both internal and external motivations when assessing technology acceptance.

- (3) SD: TAM still has shortcomings in evaluating aspects such as system design and the interaction between users and technology [22]. Nielsen [38] believes that the overall impression a system presents is the sum of various considerations, with a focus on the system's usefulness. Therefore, incorporating SD as a variable is beneficial for a holistic evaluation of an AR tour system's effectiveness.
- (4) BI: As a strong predictor of actual user behavior, BI is influenced by factors such as PU, PEOU, and SD. Because BI serves as a reliable proxy for actual usage, it is considered a primary dependent variable in technology acceptance research [39,40,41]. Consequently, evaluating the factors that shape BI can provide effective and predictive insights into a technology's potential adoption [42].

3. Methodology

3.1. Research sample and experimental procedure

The participants in this study were tourists visiting the Chiayi Sawmill Park. A total of 80 visitors were recruited to participate in the study. Before entering the Chiayi Sawmill, tourists were provided with information about the research project and its objectives. Upon agreeing to participate in the study, they signed a consent form and were informed that they could withdraw from the study at any time. At the park entrance, participants installed the AR tour app. They then used the app to scan QR codes located on various outdoor buildings and environments to experience the AR tour. Upon completing the AR tour, participants were asked to fill out a questionnaire at the exit.

The Chiayi Sawmill is a historical industrial site of significant value in Taiwan, built in 1914. It was the largest lumber industry park in the country at that time, equipped with the most advanced facilities and technologies, responsible for storing and processing wood harvested from Alishan. Although the Chiayi Sawmill no longer operates as a sawmill, it preserves many historical buildings and relics. The tour system combines AR technology and animations, allowing tourists to learn about the process of turning logs into lumber through their mobile devices. With engaging storytelling and detailed 3D architectural images, the tour system effectively transports visitors back in time, providing an in-depth understanding of the sawmill's historical context. In addition to narrative tours, the system offers various modes, including AR and textual tours. It features ten outdoor cultural heritage sites with corresponding AR detection panels (Fig. 2), enabling tourists to scan the panels and recreate the historical appearance of the sawmill. Fig. 3

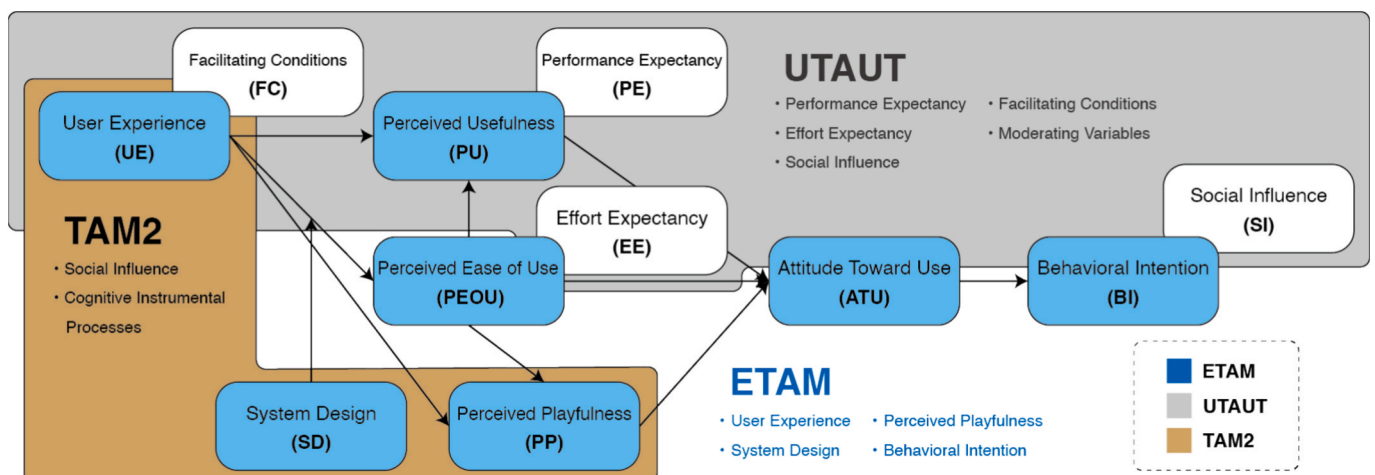


Fig. 1. Characteristics Comprising the Extended Technology Acceptance.

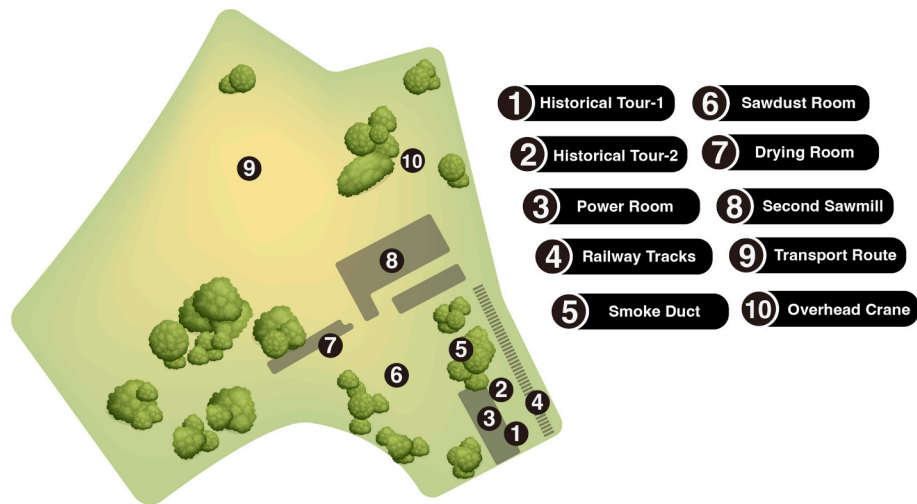


Fig. 2. 10 AR detection points of Chiayi Sawmill.

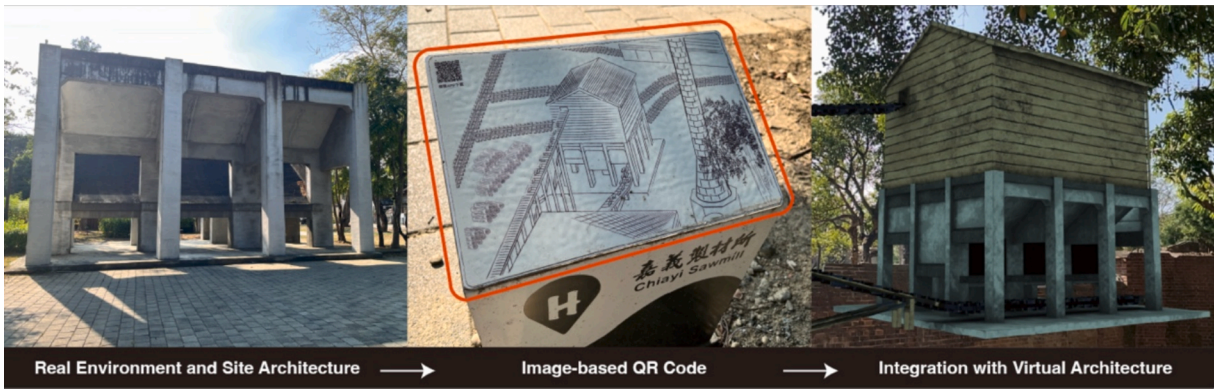


Fig. 3. The outdoor architectural ruins of Chiayi Sawmill.

shows the original site of the Chiayi Sawmill’s Sawdust Room, where scanning the AR detection panel reveals the building’s original appearance and animations with audio descriptions of the wood transportation process.

3.2. Questionnaire design and analysis

After conducting reliability testing on the measurement tools, this study retained 25 questions, which were evaluated using a Likert scale to assess tourists’ acceptance of the AR tour and their attitudes and perceptions of various variables. The Cronbach’s alpha was 0.901, ensuring the high quality and validity of the measurement tools, and addressing the ten research hypotheses proposed by ETAM. The questionnaire results will be analyzed using regression analysis to explore the direct and indirect relationships among the variables and effectively clarify the differences in technology acceptance as proposed by ETAM.

4. Results

4.1. Demographic analysis

A total of 80 visitors from the Chiayi Sawmill participated in this study. The detailed demographic characteristics of the participants are presented in Table 1, providing an essential context for the subsequent analysis of technology acceptance.

The sample was predominantly female, with 54 participants (67.5 %) compared to 26 male participants (32.5 %). The age distribution was

Table 1 Demographic Characteristics of Participants (N = 80).			
Variable	Category	Frequency (n = 80)	Percentage (%)
Gender	Male	26	32.5 %
	Female	54	67.5 %
Age Group	< 20 years	7	8.8 %
	21–30 years	11	13.8 %
	31–40 years	26	32.5 %
	41–50 years	16	20.0 %
	51–60 years	16	20.0 %
	61–70 years	4	5.0 %
Education Level	High school or below	12	15.0 %
	Undergraduate degree	39	48.8 %
	Postgraduate degree or higher	29	36.3 %
Previous Digital Tour Experience	No	19	23.8 %
	Yes	61	76.3 %
Daily Mobile Device Usage	Never	6	7.5 %
	Occasionally	13	16.3 %
	Often	14	17.5 %
	Daily	47	58.8 %

Note. N = Total number of participants. Percentages may not total 100 % due to rounding.

broad, with the largest cohort being the 31–40 age group (32.5 %, $n = 26$). Notably, participants aged between 31 and 60 collectively represented a significant portion of the sample (72.5 %, $n = 58$), indicating that the primary users were young to middle-aged adults.

The participants exhibited a high level of education. A vast majority held either a university degree (48.8 %, $n = 39$) or a postgraduate degree (36.3 %, $n = 29$). This high level of education may suggest that participants had greater expectations regarding the informational value and usefulness (Perceived Usefulness) of the tour content.

Regarding technology familiarity, the data indicates a high level of digital literacy among participants. Over three-quarters of the sample (76.3 %, $n = 61$) reported previous experience with digital tours. Furthermore, a similar proportion of participants were frequent users of mobile devices, with 76.3 % ($n = 61$) reporting they use their devices “often” or “daily”.

The demographic data reveals a participant sample of well-educated, tech-savvy adults across a mature age range. While these participants were familiar with general mobile technology and digital tours, their specific experience with AR was not detailed, positioning them as an ideal group to assess the acceptance of a novel AR system. This context is crucial for interpreting the study’s findings, particularly the significant roles that PEOU and PP play in shaping user intentions.

4.2. Descriptive statistical analysis of each dimension

To thoroughly explore the acceptance of AR tour systems in outdoor cultural heritage settings, this study conducted detailed descriptive statistical analyses on seven main dimensions: UE, SD, PU, PEOU, PP, ATU, and BI. The analysis results showed that on a five-point Likert scale, the average scores for all dimensions exceeded 4.0. This not only indicates that respondents had a consistently positive reaction to the questions but also suggests a positive attitude and perception towards the AR tour system.

Additionally, the results from the Pearson correlation coefficient matrix (Table 2) showed that most dimensions had a significant positive impact on attitudes towards using the AR tour system. Among these, UE and SD were highly correlated with both PU and ATU; PU was also highly correlated with ATU. It was also observed that PP had a high correlation with BI, being the highest among all correlation values, although it was moderately correlated with all other variables. PEOU was the only dimension that did not show a high correlation with any other variable.

A further detailed analysis of the results for each hypothesis (Fig. 4) reveals the relationships between UE, PEOU, PP, ATU, and BI in the AR tour system and attempts to explain the reasons behind these relationships (Table 3).

The analysis of the structural model reveals several key relationships that shape user acceptance of the AR tour system.

(1) The Foundational Role of User Experience and System Design

First, UE emerged as a critical antecedent. Consistent with the

literature, UE demonstrated a significant positive impact on both PU ($\beta = 0.456$, $p < 0.001$) and PEOU ($\beta = 0.377$, $p < 0.001$). This confirms that a positive initial experience is fundamental to shaping users’ perceptions of the system’s utility and usability.

Furthermore, the impact of UE on PEOU was significantly moderated by SD ($\beta = 0.234$, $p < 0.001$). As illustrated in Fig. 5, a well-designed system amplifies the positive effect of prior experience, enhancing the perceived ease of use of the AR tour. This addresses calls by Nielsen [38] and Venkatesh and Davis [22] to consider design aspects, underscoring its practical importance in enhancing both UE and PEOU.

(2) The Interplay of Perceptual Variables: Usefulness, Ease of Use, and Playfulness

The relationships between the core perceptual variables yielded nuanced findings. While PU was a strong predictor of ATU ($\beta = 0.556$, $p < 0.001$), aligning with Yen (2023), the direct influence of PEOU on PU was not statistically significant ($t = 2.72$). This contrasts with classic models like TAM and suggests that the link between ease of use and usefulness might be more complex in an AR context, possibly moderated by other factors.

Interestingly, while UE did not directly impact PP ($p = 0.187$), PEOU had a significant positive effect on it ($\beta = 0.454$, $p < 0.001$). This suggests that system usability, rather than a user’s prior experience, is the key to unlocking the playfulness aspect of the AR tour. This finding supports arguments that an enjoyable and effortless process enhances playfulness.

(3) The Primary Drivers of Attitude and Behavioral Intention

A pivotal finding concerns the determinants of user attitudes and intentions. For ATU, only PU was a significant factor. The effects of PEOU ($p = 0.744$) and PP ($p = 0.820$) were not significant. This indicates that user attitudes toward the AR tour (as measured by ATU) are primarily shaped by its perceived utility and informational value, a finding consistent with studies suggesting tourists prioritize content usefulness over entertainment in AR heritage tours. Most notably, the drivers of BI diverged from ATU. The impact of ATU on BI was not significant ($p = 0.361$), failing to support hypothesis H10. Instead, BI was most powerfully predicted by PP ($\beta = 0.672$, $p < 0.001$) and, to a lesser extent, PEOU ($\beta = 0.231$, $p < 0.001$). This core finding strongly suggests that while ATU is formed based on its utility, the ultimate intention to use the system is driven by factors of playfulness and ease of use.

In addition to these significant findings, several hypothesized paths were not supported, namely the effects of ATU on BI, PP on ATU, PEOU on ATU, and UE on PP. These non-significant results provide important theoretical insights. First, the lack of a significant ATU–BI path suggests that in outdoor cultural heritage tours, behavioral intention is more directly influenced by playfulness and usability rather than by overall attitudes toward the system. Second, although PP strongly predicted BI, it did not significantly shape ATU. This indicates that while visitors may

Table 2
Pearson’s Correlation Coefficients.

	UE	SD	PU	PEOU	PP	ATU	BI
UE	1						
SD	0.691**	1					
PU	0.707**	0.751**	1				
PEOU	0.474**	0.432**	0.519**	1			
PP	0.368**	0.337**	0.550**	0.542**	1		
ATU	0.737**	0.785**	0.763**	0.453**	0.405**	1	
BI	0.367**	0.382**	0.585**	0.655**	0.880**	0.402**	1

Note. ** Correlation is significant at the 0.01 level (2-tailed).

User Experience (UE), System Design (SD), Perceived Usefulness (PU), Perceived Ease of Use (PEOU), Perceived Playfulness (PP), Attitude Toward Use (ATU), Behavioral Intention (BI).

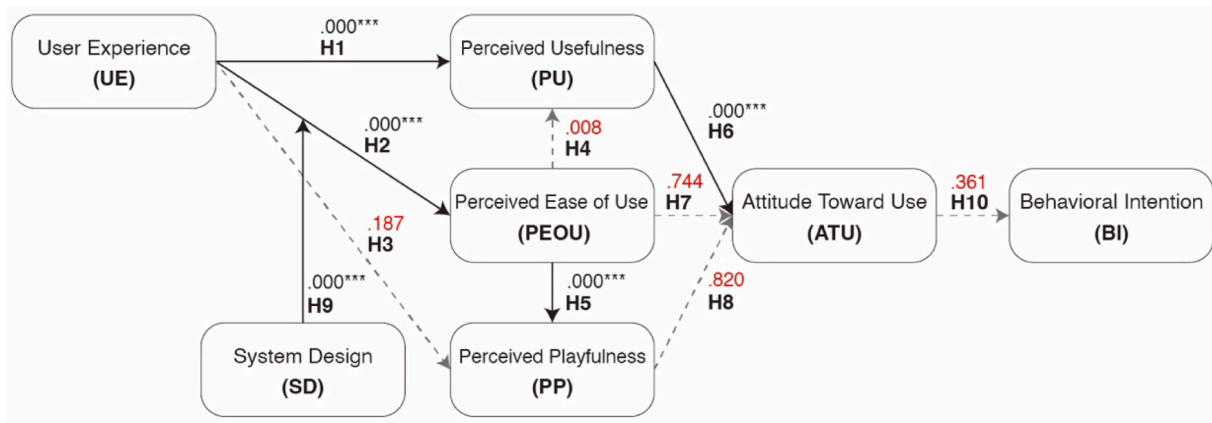


Fig. 4. Extended Technology Acceptance Model.

Table 3
Results of the Structural Model.

H	Path	B	SE	β	t	p	Supported
H1	UE → PU	0.456	0.067	0.595	6.81	0.000***	Yes
H2	UE → PEOU	0.377	0.079	0.474	4.76	0.000***	Yes
H3	UE → PP	0.109	0.082	0.143	1.33	0.187	No
H4	PEOU → PU	0.229	0.084	0.237	2.72	0.008	No
H5	PEOU → PP	0.454	0.103	0.474	4.41	0.000***	Yes
H6	PU → ATU	0.556	0.123	0.485	4.54	0.000***	Yes
H7	PEOU → ATU	0.031	0.095	0.028	0.33	0.744	No
H8	PP → ATU	-0.023	0.100	-0.020	-0.23	0.820	No
H9	SD → UE & PEOU	0.234	0.045	0.525	5.16	0.000***	Yes
H10	ATU → BI	-0.063	0.068	-0.076	-0.92	0.361	No

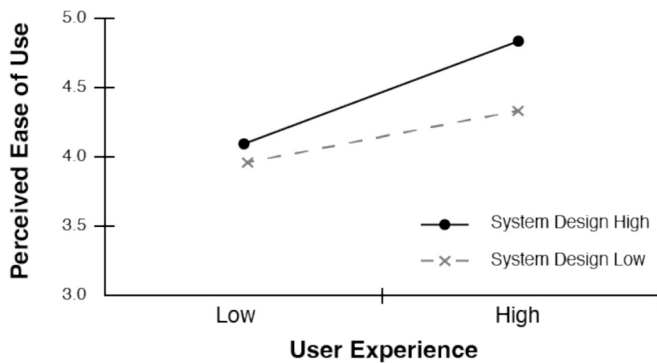


Fig. 5. Moderating Effect of System Design on User Experience and Perceived Ease of Use.

find the AR tour enjoyable, such enjoyment does not necessarily alter their general attitudes toward the system. Third, the absence of a significant effect of PEOU on ATU may be explained by the characteristics of the sample, which consisted largely of highly educated and digitally literate participants. For these users, ease of use is regarded as a baseline expectation rather than a determinant of attitude. Finally, the non-significant UE-PP path suggests that prior digital experience does not automatically enhance PP; instead, the intuitiveness and fluency of the system design play a more decisive role. Taken together, these findings highlight the necessity of adapting ETAM to cultural heritage contexts, indicating that attitude is not always the primary mediator and that playfulness and ease of use can directly drive behavioral intention.

5. Discussion

5.1. Comparison with previous research

This study validated the applicability of the ETAM in outdoor cultural heritage tours and revealed several notable departures from TAM and UTAUT findings. First, PP emerged as the strongest determinant of BI, while ATU was not significant. This challenges the TAM assumption that attitudes mediate intention [20] and aligns with prior research highlighting the importance of playfulness in immersive applications [37]. Second, UE was found to significantly enhance PU and PEOU, but not PP. This contrasts with earlier studies emphasizing the entertainment-enhancing role of prior experience [34], suggesting that SD is the more decisive factor for playfulness in heritage contexts. Finally, the non-significant effect of PEOU on ATU differs from traditional TAM results [22], possibly due to the high digital literacy of the sample, where usability is regarded as a baseline expectation rather than an attitude-shaping factor.

5.2. Explanation and observations on non-significant findings

Several hypothesized paths—ATU to BI, PP to ATU, PEOU to ATU, and UE to PP—were not supported, offering important observations. First, the non-significant ATU to BI path suggests that in outdoor cultural heritage contexts, visitors' BI are more directly shaped by PP and PEOU, rather than mediated by ATU. Second, the null PP to ATU result indicates that finding the system “enjoyable” does not necessarily change overall attitudes, although playfulness still directly drives behavioral intention. Third, the lack of a significant PEOU to ATU relationship implies that usability is treated as a baseline requirement rather than a key factor in shaping attitudes. Finally, the absence of a UE to PP link underscores the critical role of intuitive and seamless system design in fostering playfulness. These observations suggest that in experience-driven contexts, the ETAM should place greater emphasis on the direct

effects of PP and PEOU on BI, while reconsidering the mediating role of attitudes.

5.3. Practical implications

The findings of this study provide several actionable implications for the design and implementation of AR-based cultural heritage tours. First, system design should not remain at the level of simply digitizing existing content. Instead, it should integrate exploratory and immersive interaction or game mechanisms aligned with the heritage theme to enhance PP and thereby strengthen BI. Such an approach enables visitors to transition from passive recipients of information to active participants, enriching their experience with engagement and enjoyment. The interface and navigation processes should be kept simple and intuitive, supported by clear guidance to minimize operational complexity. This ensures that visitors can remain immersed in the content rather than being distracted by technical hurdles. Particular attention should be given to older adults and users with lower digital literacy, for whom reducing barriers is crucial to preventing exclusion caused by technological divides.

In addition, differentiated design strategies should be adopted for heritage sites of varying types and scales. For large-scale archaeological sites or extensive cultural landscapes, system stability, positioning accuracy, and the potential of markerless AR are critical. However, increasingly sophisticated AR content also places heavier demands on personal mobile devices, which may reduce usability and satisfaction. Providing dedicated devices on-site can encourage adoption and minimize issues related to downloading, installation, and compatibility. For smaller venues or single-site ruins, greater emphasis may be placed on personalized storytelling and refined narrative content to deepen visitor immersion. Meanwhile, urban museums and modern exhibition halls are well suited to integrating advanced interactive technologies—such as projection mapping, sensor-based installations, and mixed-reality elements—to deliver a more refined and immersive digital tour experience that enhances both engagement and cultural learning.

Finally, AR tour design must strike a balance between educational and entertainment values. Excessive gamification risks diluting cultural meaning. Instead, contextualized narratives and well-calibrated interactive elements should be employed to sustain engagement while simultaneously promoting cultural learning outcomes.

5.4. Limitations and future research

This study is subject to several limitations. First, it focused on a single case—the Chiayi Sawmill—so the findings may not be directly generalizable to other types or scales of cultural heritage sites, and the external validity requires further testing. Second, the sample consisted mainly of highly educated and digitally literate participants, limiting representation of diverse user groups such as older adults or digitally disadvantaged populations, whose patterns of acceptance may differ. Third, the study employed a one-time intervention design without longitudinal tracking or cross-context comparisons, preventing assessment of the robustness of AR tours in long-term use or varied settings. Finally, the study did not include cross-cultural comparisons; cultural contexts may shape the relative importance of factors such as PU and PP. Future research should therefore adopt multi-site, more diverse, and cross-cultural designs to further validate the applicability of the ETAM and clarify how cultural and demographic characteristics influence technology acceptance.

6. Conclusion

This study employed the Chiayi Sawmill as a case to investigate the application of AR in outdoor cultural heritage tours and to construct the ETAM for examining the key factors influencing user acceptance. The findings addressed three core research questions: (1) whether ETAM can

effectively explain visitors' BI (2) whether AR technology enhances the experiential value of cultural heritage tours; and (3) whether user acceptance is primarily driven by PP, UE, and SD.

Three major conclusions were reached. First, PP plays a decisive role in shaping BI, indicating that visitors to cultural heritage tours place greater emphasis on interactivity and enjoyment. Second, UE and SD significantly enhance PU and PEOU, thereby facilitating acceptance. Third, the relatively limited effects of PEOU and ATU on BI suggest that in contexts characterized by widespread digital literacy, visitors focus more on whether the system successfully integrates both educational and entertainment values.

From an academic perspective, this study extends TAM2 and UTAUT by incorporating PP, UE, and SD, thereby proposing a model better suited to outdoor cultural heritage contexts and addressing the limitations of existing frameworks in immersive applications. From a practical perspective, the findings provide valuable insights for heritage managers and system designers, highlighting the importance of balancing educational and entertainment content in digital heritage tours to enhance visitor engagement and cultural learning outcomes.

CRedit authorship contribution statement

Yi-Chun Li: Writing – original draft, Methodology, Investigation, Data curation. **Wen-Huei Chou:** Writing – review & editing, Project administration, Conceptualization. **Yen-Liang Wu:** Visualization, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.entcom.2025.101034>.

Data availability

Data will be made available on request.

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